

NODE.js

Interview Questions from Beginner to Advance

1. Node.js Basics

- What is Node.js, and how does it work internally?
- Why is Node.js single-threaded, and how does it handle concurrency?
- What is the Event Loop, and how does it work in Node.js?
- What is the difference between `setImmediate()`, `process.nextTick()`, and `setTimeout()`?
- What is the role of the V8 engine in Node.js?
- What is the difference between synchronous and asynchronous programming?
- What are the advantages and disadvantages of using Node.js?

2. Modules and NPM

- What are the different types of modules in Node.js? (Built-in, Custom, Third-party)
- How does the `require()` function work in Node.js?
- What is the difference between `require()` and `import` in Node.js?
- What is the `package.json` file, and why is it important?
- What is the difference between `dependencies` and `devDependencies`?
- What is `npm`, and how is it different from `npm`?
- What are CommonJS and ES Modules, and how do they differ?

3. File System and Streams

- How do you read and write files in Node.js?
- What is the difference between `fs.readFile()` and `fs.createReadStream()`?
- What are streams in Node.js, and what are their types? (Readable, Writable, Duplex, Transform)
- What is a buffer in Node.js, and how is it used?
- How do you handle large file uploads efficiently in Node.js?

4. Express.js and Middleware

- What is Express.js, and why is it used in Node.js applications?
- What are middleware functions in Express?
- What is the difference between global and route-specific middleware?
- How do you handle errors in Express.js?
- What is the difference between `app.use()` and `app.get()`?
- How do you implement authentication in an Express app? (JWT, Sessions)
- What is CORS, and how do you enable it in Express?

5. Asynchronous JavaScript in Node.js

- What are callbacks in Node.js, and what are their drawbacks?
- What are Promises, and how do they work in Node.js?
- What is async/await, and how does it improve asynchronous code?
- What is the difference between callbacks, Promises, and async/await?
- How do you handle errors in async/await functions? (try/catch, .catch())

6. Database Connectivity

- How do you connect Node.js with MongoDB using Mongoose?
- How do you perform CRUD operations using Mongoose?
- What is the difference between SQL and NoSQL databases?
- What is an ORM, and why is it used? (Example: Sequelize for SQL, Mongoose for MongoDB)
- How does connection pooling work in Node.js databases?

7. Security and Performance Optimization

- How do you prevent SQL Injection in Node.js?
- What are some security best practices for a Node.js application?
- How do you prevent memory leaks in a Node.js application?
- What are the best practices for optimizing performance in Node.js?
- What is rate limiting, and how do you implement it in Node.js?
- What is Helmet.js, and how does it improve security?

8. API Development and Testing

- What is RESTful API, and how do you create one using Express?
- What is GraphQL, and how is it different from REST?
- How do you handle file uploads in an API using Node.js?
- How do you write unit tests for a Node.js application?
- What is the difference between Mocha, Jest, and Chai for testing?
- What is Postman, and how is it used for API testing?

9. Process Management and Deployment

- How do you handle environment variables in Node.js?
- What is PM2, and how does it help in process management?
- How do you deploy a Node.js application? (Heroku, AWS, Vercel, DigitalOcean)
- What is Docker, and how do you use it with a Node.js application?
- How do you scale a Node.js application?
- What is load balancing, and how do you implement it in Node.js?

10. Advanced Topics

- What is WebSockets, and how do you use them in Node.js?
- What is Server-Sent Events (SSE), and how does it differ from WebSockets?
- How does clustering work in Node.js?
- What are worker_threads in Node.js, and when should you use them?
- What is gRPC, and how does it compare with REST and GraphQL?
- What are microservices, and how do you implement them in Node.js?
- What is the difference between monolithic and microservices architecture?
- What is an Event-Driven Architecture, and how does it work in Node.js?

11. Real-World Scenario-Based Questions

- If your Node.js app is running slow, what steps would you take to diagnose and fix the issue?
- How would you implement authentication and authorization in a Node.js API?
- How do you handle millions of concurrent requests in a Node.js application?
- How would you design a chat application using Node.js and WebSockets?
- How would you build a rate limiter for an API in Node.js?